

Build AI Agents in 21 Days

Your roadmap, your stack, and how to study. Read this once before Day 1.

What you will have at the end

A working AI research agent that you built yourself, running as a real service, with a dashboard in front of it. Not a tutorial copy. Your own code, which you can explain line by line and change without fear.

More importantly: you will be able to build things that are not in any tutorial, because you will understand what is happening underneath.

The big idea (read this twice)

Twenty-one topics looks like a mountain. It is not. These are really **six ideas**, taught three times each, a little deeper every time.

Idea	What it means in plain words	Days
The call	Send text to a model, get text back	1
State	Remembering what happened before	2, 11
Instruction	Controlling behaviour using words	3
Tools	Letting the model trigger your own functions	4, 8, 13
Retrieval	Giving the model knowledge it does not have	5, 6, 12
The loop	Model decides, acts, looks at the result, repeats	7, 9, 10, 14

Everything after Day 14 is packaging: making it reliable (15), using someone else's version of these six (16 to 19), putting it online (20), and joining it all together (21).

An AI agent is a loop, plus a model, plus tools, plus memory, plus a rule for when to stop.

That is the whole secret. The 21 days simply make each of those five words rich.

The 21 days

Five modules. Each day has one real question it answers.

01 - BUILD THE FOUNDATION Days 1 to 5

Day	Topic	The question it answers
1	Build Your First AI Chatbot	How do I talk to a model from my own code?
2	Conversation Memory	Why does it forget, and what do I send back?
3	Prompt Engineering	How do I make behaviour reliable, not lucky?
4	AI Tools	How does a text model actually do things?
5	Document AI: Read Files	How do I get my own data into the prompt?

02 - AI AGENTS Days 6 to 9

Day	Topic	The question it answers
6	Build Your First RAG System	What if my data is too big for the prompt?
7	Build Your First AI Agent	Who decides the next step, me or the model?
8	Model Context Protocol (MCP)	How do tools become reusable everywhere?
9	Autonomous AI Agent	How does it run without me approving each step?

03 - ADVANCED AGENT SYSTEMS Days 10 to 15

Day	Topic	The question it answers
10	Advanced Agent Planning	How does it break a big goal into steps?
11	Memory Engineering	What should it remember, and what should it forget?
12	Advanced RAG	Why is it finding the wrong information?
13	MCP in Depth	How do I build my own tool server, safely?
14	Multi-Agent Systems and Orchestration	When are many agents better than one?
15	Making AI Agent Production Ready	Cost, speed, failures, limits, testing.

04 - FRAMEWORKS Days 16 to 19

Day	Topic	The question it answers
16	LangGraph	Agents drawn as a map of states.
17	CrewAI	Agents as roles in a team.
18	Build Our AI Research Agent	A full build using a framework.
19	LangChain for Agent Engineers	The glue layer, and when to avoid it.

05 - DEPLOYMENT AND CAPSTONE Days 20 to 21

Day	Topic	The question it answers
20	AI Agent Deployment	Putting it online so other people can use it.
21	Capstone: Research AI Agent Platform	Build the whole thing. Ship it.

One project, twenty-one upgrades

You will not build 21 small throwaway scripts. You build **one** thing and add to it every day: a **Research Assistant**. By Day 21 the capstone is already most of the way built.

After day	Your assistant can...
Day 1	Reply to you
Day 2	Remember the conversation
Day 3	Follow rules and behave consistently
Day 4	Run your own Python functions
Day 5	Read a file you give it
Day 6	Search across many documents
Day 7	Choose its own next step
Day 9	Work alone, without you approving each step
Day 11	Remember things across days, not just one chat
Day 14	Split work between several agents
Day 20	Run online as a real service
Day 21	Be a platform, with your dashboard in front

Your stack

Two languages, two jobs. This is also how real teams build, so you are not making a compromise.

Part	Choice	Why
AI code	Python	All the AI libraries live here
Dashboard, UI	Node.js	Better at web things
They talk by	HTTP	Python is the brain, Node is the face
Model	Groq free tier, llama-3.3-70b-versatile	Free, no card, fast, does tool calling
Backup model	Gemini free tier	For when Groq limits you
Embeddings	sentence-transformers, on your laptop	Free and unlimited, no API needed
Vector store	Chroma, local	No server, no account
Python server	FastAPI	Needed from Day 20

One decision worth understanding. All model calls are written using the OpenAI-compatible client. Groq, Gemini and models running on your own laptop all accept that same shape. So changing provider means changing two lines, never rewriting your code.

One warning. The Groq free tier allows roughly 30 requests and 6,000 tokens per minute. The token limit is the one that bites. From Day 9 your agent sends the whole conversation on every loop. That is why retry-with-waiting goes into your code on Day 1, not Day 15.

Before you start Day 1

You do not need to know AI. You do need these. If two or more are shaky, spend one day on them first. It will save you a week of confusion.

- Python: functions, lists, dictionaries, loops
- Running **pip install** and knowing what a virtual environment is
- Reading a Python error message without panic
- Using a terminal: moving between folders, running a file
- Basic JavaScript, for the dashboard side only (needed from Day 20)

Also: a laptop with about 4 GB of free space, and a free Groq account from console.groq.com. No credit card is needed anywhere in this course.

How to study each day

Five stages, always in this order. Most people stop after stage two. The learning is in stage three.

Stage	Time	What it means
1. READ	20 min	Read the lesson once. Do not type anything yet.
2. CODE	45 min	Type it yourself. Never copy-paste. Typing forces you to read.
3. EXPERIMENT	45 min	Break it on purpose. This is the real class.
4. UNDERSTAND	20 min	Answer the check questions out loud, without looking.
5. BUILD	30 min	Make one small change that is yours, not the lesson's.

Short on time? Read 20, Code 30, Experiment 30. Ninety minutes is enough. Never skip Experiment. Cutting that is cutting the course.

The six rules

1. **Type the code.** Copy-paste teaches your fingers nothing and your head less.
2. **Print the raw thing.** The actual prompt sent, the actual tool call, the actual text found. Frameworks hide these. People who can debug agents are the people who still look underneath.
3. **Break something every day.** Change a number to an extreme. Delete a line that looks important. Read the error properly.
4. **Do not skip ahead to the frameworks.** Days 16 to 19 will make Days 7 to 14 look like wasted time. They are not. Learn the framework first and you will never be able to fix it, because you will not know what it is doing.
5. **Keep one notes file.** After each day, paste in the PROJECT STATE block. That is how the next day knows what you already built.
6. **Explain it to someone.** Out loud, no notes. If you cannot, you have not learned it yet. That is useful information, not failure.

Checkpoints: how to know it is working

At these points, you should be able to do the following with no notes in front of you.

After	You should be able to...
Day 5	Explain why the model forgets, and draw what is actually sent on message number ten of a conversation.
Day 9	Draw the agent loop on paper and name its stopping rule. Say when retrieval is the wrong answer.
Day 15	Say why your agent is slow and expensive, and name the three things you would change first.
Day 19	Read framework code and say what it is doing underneath.
Day 21	Build something that is not in this syllabus, without a tutorial.

If you fail a checkpoint, stop and go back. Falling behind the schedule costs you a few days. Falling behind the understanding costs you the whole course.

Now open Day 1. The first thing you build takes eight lines of code.